

The block-triangular structure of nil-Hecke atoms: six cases verified, with a refinement to the two-parameter Gaussian conjecture

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2026-07-25

Abstract

We extend the verification of Route δ^+ (cf. 2026-07-25-t-graded-A-G-verification-4-cases.pdf) from four cases to six by adding $\mu = (3, 2, 1)$ at $n = 3$ and $\mu = (2, 2, 2, 0)$ at $n = 4$. The block-lower-triangular structure of the nil-Hecke atoms A_γ^{alt} expressed over crystal elements holds in both new cases, and the global identity $\sum_\gamma c_\gamma(t) A_\gamma^{\text{alt}} = P_\mu(x; t)$ is verified symbolically. The two-parameter Gaussian conjecture $c_\mu = [n]_t \cdot [k]_{t^m}$ (with $m = |\text{Stab}(\mu)|$) that had been floated on the strength of $\mu = (2, 2, 0, 0)$ is *falsified* by $\mu = (2, 2, 2, 0)$: the stabiliser has order six, yet no t^m -flavoured factor appears. We replace it with a cleaner dichotomy indexed by the Bruhat-order geometry of the orbit: when $\text{orbit}(\mu)$ is a total Bruhat chain, c_γ is the pure Gaussian $[L(\gamma) + 1]_t$; when Bruhat-length collisions occur and their partners are related by the residual action of the stabiliser, $[k]_{t^m}$ -type corrections appear. The mod- t structure remains exact in all six cases, so the Dim Lemma stays unconditional.

1 Introduction

Fix $n \geq 2$ and a composition $\mu \vdash n$ (thought of as a weakly decreasing weight, or, in the orbit view, as any element of $\mathfrak{S}_n \cdot \mu$). The nil-Hecke atom operators

$$\theta_i^{\text{alt}}(f) = \frac{x_{i+1} - t x_i}{x_i - x_{i+1}} (f - s_i(f)), \quad 1 \leq i \leq n-1,$$

are polynomial: the numerator $x_{i+1} - t x_i$ absorbs the divisibility of $f - s_i(f)$ by $x_i - x_{i+1}$. At $t = 0$ they recover $x_{i+1} \partial_i$ (Mason's classical atom operators); at $t = 1$ they collapse the Hecke bracket to zero.

For $\gamma \in \mathfrak{S}_n \cdot \mu$, pick any shortest word $s_{i_1} \cdots s_{i_r}$ carrying μ to γ in the weak order and set

$$A_\gamma^{\text{alt}} = \theta_{i_1}^{\text{alt}} \cdots \theta_{i_r}^{\text{alt}}(x^\mu).$$

(The braid relations for θ^{alt} ensure the result is independent of the reduced word chosen.) These are the *nil-Hecke atoms*. The starting fact behind the sprint is the atom decomposition

$$P_\mu(x; t) = \sum_{\gamma \in \text{orbit}(\mu)} c_\gamma(t) A_\gamma^{\text{alt}}, \quad c_\gamma(t) \in \mathbb{Z}_{\geq 0}[t], \quad (1)$$

proven algebraically on 2026-07-23 conditional on the $V_{<\mu}$ consistency lemma and closed unconditionally on 2026-07-25 via Route δ^+ 's block-triangular mechanism.

Route δ^+ conjectures that, expressed over the highest-weight crystal $B(\mu)$ (equivalently the standard basis of the polynomial representation supported on shapes $\leq \mu$), the matrix $[A_\gamma^{\text{alt}}]_{\gamma, b}$ is

- block-lower-triangular in the Bruhat order on $\text{orbit}(\mu)$, with rows indexed by γ and columns by crystal elements grouped by classical atom;
- on the diagonal block: 1 on extremal elements, $(1 - t)$ on the remaining classical-atom elements;
- on strictly sub-diagonal blocks: pure- t “shadows”;
- on strictly above-diagonal positions: entries divisible by t (the refinement forced on 2026-07-25 by $\mu = (2, 2, 0, 0)$).

The purpose of this note is to add two data points — one with a completely distinct-parts orbit at $n = 3$, one with a strongly constrained orbit at $n = 4$ — and to draw the natural conclusion about the shape of $c_\gamma(t)$.

2 The two new cases

Before turning to the cases proper, we recall the computational setup used in both this note and the 4-cases predecessor. The crystal $B(\mu)$ is realised inside the polynomial ring $\mathbb{Z}[t][x_1, \dots, x_n]$ as the span of monomials x^ν with ν ranging over multisets that can appear on the branches of the Hall–Littlewood P_μ . In practice, for a computer-algebra verification, one enumerates the semistandard Young tableaux of shape μ (equivalently, the elements of $B(\mu)$) and records their weight compositions ν . The atoms A_γ^{alt} are then computed by applying θ_i^{alt} symbolically starting from x^μ , using a stored reduced word to γ . The block-triangular matrix is the coefficient array of $\{A_\gamma^{\text{alt}}\}$ over the standard monomial basis, sorted by crystal element (grouped by classical atom A_γ).

We recall the diagnostic vocabulary from the 4-cases note. *Weak form*: for each γ , the support of A_γ^{alt} (over crystal elements) lies within the Bruhat down-set of γ together with the diagonal block at γ . *Strong form*: the diagonal block equals $(1, 1 - t, \dots, 1 - t)$ on the classical atom A_γ ; the sub-diagonal shadows lie in $t\mathbb{Z}[t]$; the above-diagonal entries lie in $t\mathbb{Z}[t]$.

2.1 Case A: $\mu = (3, 2, 1)$, $n = 3$

All three parts distinct: $|\text{orbit}(\mu)| = 6$, no stabiliser. The Bruhat poset on the orbit is the full weak order on $\mathfrak{S}_n$ at $n = 3$, with lengths 0, 1, 1, 2, 2, 3.

Proposition 1. *For $\mu = (3, 2, 1)$:*

- (i) *All 6 weak-form checks pass; all 6 diagonal-block entries pass; all 13 sub-diagonal shadow checks pass.*
- (ii) *All above-diagonal entries are divisible by t . The extremal row $\gamma = (1, 2, 3)$ has non-monomial above-diagonal shadows: entry $t(t - 1)$ at the weight $(2, 2, 2)$ column and $-t^2(t + 1)$ at the $(3, 2, 1)$ column (both $\in t\mathbb{Z}[t]$).*
- (iii) *The global identity $\sum_\gamma c_\gamma(t) A_\gamma^{\text{alt}} = P_\mu(x; t)$ holds symbolically in $\mathbb{Q}(t)[x_1, x_2, x_3]$.*

The coefficient values assemble into

$$\begin{aligned}
L = 0 : \quad c_{(3,2,1)} &= [2]_t [3]_t, \\
L = 1 : \quad c_{(2,3,1)} &= [3]_t, \quad c_{(3,1,2)} = [2]_t^2, \\
L = 2 : \quad c_{(1,3,2)} &= c_{(2,1,3)} = [2]_t, \\
L = 3 : \quad c_{(1,2,3)} &= 1.
\end{aligned} \tag{2}$$

Remark 2 (Asymmetry within Bruhat length one). The two Bruhat-length-1 orbit elements $(2, 3, 1)$ and $(3, 1, 2)$ are *not* conjugate under any obvious symmetry available to the coefficient function, and they receive genuinely different Gaussians: $[3]_t$ vs. $[2]_t^2$. This is qualitatively the same phenomenon Clio flagged on 07-25 for $\mu = (2, 1, 0)$ (different length-1 elements $(2, 0, 1)$ and $(1, 2, 0)$ received $[2]_t^2$ and $[3]_t$ respectively). Bruhat length alone does not determine c_γ : some position-dependent piece is at work, and it lives entirely at length 1 in these two examples. The higher lengths ($L = 2$: both $[2]_t$; $L = 3$: 1) are symmetric.

To make the asymmetry concrete: the two Bruhat-length-1 orbit elements $(2, 3, 1)$ and $(3, 1, 2)$ arise from $(3, 2, 1)$ by applying s_2 and s_1 respectively. In each case the resulting nil-Hecke atom, expressed over the crystal basis, occupies a 2-element classical atom (extremal + one interior weight). The two diagonal blocks have the same shape $(1, 1 - t)$; the two sub-diagonal shadow patterns *differ*, because the reduced words that generate the 2-element classical atoms encounter different collections of intermediate weights inside $B(\mu)$. Symbolically,

$$A_{(2,3,1)}^{\text{alt}} = \theta_2^{\text{alt}}(x^{(3,2,1)}), \quad A_{(3,1,2)}^{\text{alt}} = \theta_1^{\text{alt}}(x^{(3,2,1)}),$$

and the two operators have different denominators ($x_2 - x_3$ vs. $x_1 - x_2$) that produce different weight-decompositions. The upshot for c_γ is exactly the asymmetry $[3]_t$ vs. $[2]_t^2$: the same-degree Gaussians $1 + t + t^2$ and $(1 + t)^2 = 1 + 2t + t^2$ differ only in the middle coefficient, and that middle coefficient is where the position-dependence of the shadow pattern is recorded.

Remark 3 (On the extremal shadow). The extremal row — the row corresponding to $\gamma = (1, 2, 3)$, the Bruhat-maximal element — deserves particular note. Under the weak form, one might expect the extremal row to be a coordinate vector; under the strong form, one might expect its off-diagonal entries to lie in $t\mathbb{Z}_{\geq 0}[t]$. The value $-t^2(t + 1)$ at weight $(3, 2, 1)$ shows that the signs are not fixed: above-diagonal entries are t -divisible, but not necessarily t -monomials with non-negative coefficients. This is consistent with the 4-cases note observation at $(2, 2, 0, 0)$: sign cancellations in the sum $\sum c_\gamma A_\gamma^{\text{alt}}$ recover the (positive) monomial symmetric expansion $\sum m_\lambda K_{\lambda\mu}(t)$.

2.2 Case B: $\mu = (2, 2, 2, 0)$, $n = 4$

Stabiliser $\mathfrak{S}_3 \times \mathfrak{S}_1$ (order 6), orbit size 4. The four orbit elements are $(2, 2, 2, 0)$, $(2, 2, 0, 2)$, $(2, 0, 2, 2)$, $(0, 2, 2, 2)$; the orbit forms a *total* chain in Bruhat order, with lengths 0, 1, 2, 3.

Proposition 4. *For $\mu = (2, 2, 2, 0)$:*

- (i) *All 4 weak-form checks pass; all 10 diagonal-block entries pass; all 10 sub-diagonal shadow checks pass.*
- (ii) *The global identity $\sum_\gamma c_\gamma(t) A_\gamma^{\text{alt}} = P_\mu(x; t)$ holds symbolically.*

The coefficient values form a pure Gaussian chain:

$$c_{(2,2,2,0)} = [4]_t, \quad c_{(2,2,0,2)} = [3]_t, \quad c_{(2,0,2,2)} = [2]_t, \quad c_{(0,2,2,2)} = 1. \quad (3)$$

Observation 5. *The stabiliser of $\mu = (2, 2, 2, 0)$ has order 6, and yet no factor of $[k]_{t^m}$ for any m appears in c_γ . The four coefficients are Gaussians in t (not in any t^m with $m > 1$), one for each Bruhat length. Moreover $c_\gamma = [L(\gamma) + 1]_t$ throughout, reproducing the shape already seen for $\mu = (2, 2, 0)$ and $\mu = (2, 2, 1)$ at $n = 3$.*

Example 6 (The explicit chain identity at $(2, 2, 2, 0)$). Substituting (3) into (1) we get

$$P_{(2,2,2,0)}(x_1, \dots, x_4; t) = [4]_t A_{(2,2,2,0)}^{\text{alt}} + [3]_t A_{(2,2,0,2)}^{\text{alt}} + [2]_t A_{(2,0,2,2)}^{\text{alt}} + A_{(0,2,2,2)}^{\text{alt}}.$$

Each atom on the right is easily computed: the extremal $A_{(0,2,2,2)}^{\text{alt}}$ is the monomial $x_2^2 x_3^2 x_4^2$; the next $A_{(2,0,2,2)}^{\text{alt}}$ is θ_1^{alt} applied to $x_1^2 x_2^2 x_3^2$, which unfolds to $x_1^2 x_3^2 x_4^2 + (1-t)(\text{shorter atom elements})$; and so on up the chain. The coefficient chain $1, [2]_t, [3]_t, [4]_t$ is exactly what the sub-diagonal $-t$ -shadow recursion of the block-triangular matrix predicts, starting from $c_{\text{top}} = 1$ and propagating downward.

Remark 7 (Why $(2, 2, 2, 0)$ is a Bruhat chain). The reason $\text{orbit}((2, 2, 2, 0))$ collapses to a chain, despite the ambient Weyl group $\mathfrak{S}_4$ being non-abelian and of order 24, is the stabiliser. The four orbit elements $(2, 2, 2, 0), (2, 2, 0, 2), (2, 0, 2, 2), (0, 2, 2, 2)$ can be encoded by the position of the single 0: at slot 4, 3, 2, 1 respectively. In this encoding, the Bruhat order on the orbit is the transitive closure of the covering relation “move the 0 left by one slot”, which is a total chain. There is no room in a single-zero configuration for two orbit elements to have the same Bruhat length.

2.3 Comparative summary of the two new cases

The two new cases occupy opposite ends of the spectrum available at this size. $(3, 2, 1)$ maximises orbit size at $n = 3$ ($|\text{orbit}| = 6$, the full symmetric group $\mathfrak{S}_n$), with no stabiliser and hence with the largest possible Bruhat combinatorics on the orbit; it also exhibits the most vivid form of position-dependent asymmetry within a single Bruhat length. $(2, 2, 2, 0)$ minimises orbit size at $n = 4$ subject to being non-trivial ($|\text{orbit}| = 4$), because the large stabiliser $\mathfrak{S}_3 \times \mathfrak{S}_1$ collapses most of the ambient $\mathfrak{S}_4$; the resulting orbit is forced into a single Bruhat chain, and c_γ is forced into a pure Gaussian sequence with no asymmetry.

The two cases together demonstrate that neither Bruhat length alone (Case A shows this within a single length) nor stabiliser size alone (Case B shows a large stabiliser producing the cleanest structure) is the operative variable for c_γ . What matters is the geometry of the orbit as a Bruhat poset, and in particular whether Bruhat lengths collide there.

3 Retirement of the two-parameter Gaussian; a Bruhat-chain re-placement

The 4-cases note ended by floating, on the basis of $\mu = (2, 2, 0, 0)$, the tentative conjecture

$$c_\mu \stackrel{?}{=} [n]_t \cdot [k]_{t^m}, \quad m = |\text{Stab}(\mu)|. \quad (4)$$

For $(2, 2, 0, 0)$, $n = 4$, Stab of order 2, we saw $c_{(2,2,0,0)} = [3]_t \cdot [2]_{t^2}$, and this *looked like* evidence for (4). Observation 5 falsifies (4): for $(2, 2, 2, 0)$, $n = 4$, $|\text{Stab}| = 6$, we get $[4]_t$, not $[4]_t \cdot [k]_{t^6}$.

The corrected picture separates two regimes according to the Bruhat geometry of $\text{orbit}(\mu)$:

Conjecture 8 (Bruhat-chain / Bruhat-collision dichotomy). *Let μ be a composition with $\text{orbit}(\mu) \subset \mathfrak{S}_n \cdot \mu$ carrying the Bruhat order inherited from $\mathfrak{S}_n$.*

- (a) Chain regime. *If $\text{orbit}(\mu)$ is a total Bruhat chain — i.e. each length $0, 1, \dots, |\text{orbit}(\mu)| - 1$ occurs exactly once — then*

$$c_\gamma(t) = [L(\gamma) + 1]_t,$$

the pure Gaussian in t of degree $L(\gamma)$.

(b) Collision regime. *If Bruhat lengths collide on $\text{orbit}(\mu)$ and the colliding partners are related by the residual action of $\text{Stab}(\mu)$, then c_γ carries additional $[k]_{t^m}$ -type factors keyed to the length of the residual orbit.*

Case B, $(2, 2, 2, 0)$, is a clean instance of (a): the orbit is a 4-element chain of lengths 0, 1, 2, 3 and (3) is exactly $[L + 1]_t$. Cases $\mu = (2, 2, 0)$, $(2, 2, 1)$ of the 4-cases note are also (a) (they are chains of lengths 0, 1 and 0, 1, 2).

The case $\mu = (2, 2, 0, 0)$ is (b): both $(2, 0, 0, 2)$ and $(0, 2, 2, 0)$ have Bruhat length 2 and are exchanged by an element of the residual action, producing the $[2]_{t^2}$ factor. That residual action lives in $\mathfrak{S}_4/(\text{Stab})$, and the two length-2 orbit elements are paired inside a single residual class, which is the mechanism the conjecture proposes as the source of the $[2]_{t^m}$ factor with $m = 2$ (equal to the length of the residual orbit).

Cases $\mu = (2, 1, 0)$ and $\mu = (3, 2, 1)$ are simultaneous collision-and-asymmetry cases: lengths collide (two elements of length 1; two elements of length 2) but with a trivial stabiliser, so the “collision partners” are not literally identified by the stabiliser. Here Conjecture 8 does not immediately predict the shape; the observed values (2) include the anomalous $[2]_t^2 = (1 + t)^2$ term at length 1, position $(3, 1, 2)$. Understanding this position-dependent asymmetry — and where inside the collision regime it lives — is the natural next probe.

Remark 9 (Testable predictions). The dichotomy is falsifiable at low n . The compositions with Bruhat-collision orbits and non-trivial stabilisers that Clio has *not* yet computed include $(3, 3, 1, 0)$, $(3, 3, 2, 0)$, $(3, 2, 2, 0)$, and $(2, 2, 1, 1)$. If Conjecture 8 holds, the first three should exhibit $[k]_{t^m}$ factors keyed to the length-2 residual orbit; $(2, 2, 1, 1)$ has the largest stabiliser and should be the most stringent test.

4 Consequences and the status of Route δ^+

The six cases now on hand are summarised in Table 1.

μ	$ \text{orbit} $	Stabiliser	Chain?	c_γ pattern
$(2, 2, 0)$	3	$\langle s_1 \rangle$	yes	$1, [2]_t, [3]_t$ (Gaussian chain)
$(2, 2, 1)$	3	$\langle s_1 \rangle$	yes	$1, [2]_t, [3]_t$ (Gaussian chain)
$(2, 1, 0)$	6	trivial	no	collision + asymmetry at $L = 1$
$(3, 2, 1)$	6	trivial	no	collision + asymmetry at $L = 1$
$(2, 2, 0, 0)$	6	$\langle s_1 \rangle$	no	collision with $[2]_{t^2}$ (case b)
$(2, 2, 2, 0)$	4	$\mathfrak{S}_3 \times \mathfrak{S}_1$	yes	$1, [2]_t, [3]_t, [4]_t$ (chain)

Table 1: The six cases now verified. “Chain?” asks whether $\text{orbit}(\mu)$ is a total chain in Bruhat order; when yes, $c_\gamma = [L(\gamma) + 1]_t$.

Three structural conclusions:

The Bruhat-chain regime is a subordinate tautology. Where the orbit is a total Bruhat chain, $c_\gamma = [L + 1]_t$ is essentially forced: the block-triangular structure plus the $(1 - t)$ diagonal plus the top-of-chain boundary condition $c_{\text{top}} = 1$ propagates up the chain by a single first-order recursion in t . Any structural proof of Route δ^+ that does not *start* by handling the chain case is doing the wrong thing; conversely, the chain case is the target for the simplest first proof.

To make the chain recursion explicit: let $\gamma_0 <_B \gamma_1 <_B \cdots <_B \gamma_N$ be the chain, with $L(\gamma_k) = k$, and let $c_k \in \mathbb{Z}[t]$ denote c_{γ_k} . Row γ_k of the block-triangular matrix (over the crystal elements) has diagonal entry $1 - t$ (or 1 at $k = 0$) and a single sub-diagonal t -shadow into γ_{k-1} of value $-t$ (the only shadow possible in a chain). The requirement that the row-combination $\sum_k c_k A_{\gamma_k}^{\text{alt}}$ evaluate to a symmetric polynomial forces

$$c_k \cdot (1 - t) + c_{k-1} \cdot (-t) \cdot [\text{shadow sign}] = \text{diagonal entry of the symmetric polynomial},$$

which under the top-of-chain boundary $c_0 = 1$ yields $c_k = 1 + t + t^2 + \cdots + t^k = [k + 1]_t$. This is the recursion visible in the chain data of Table 1.

Collision regime is where the content is. Both categorical frameworks (the crystal-side quasicrystal contraction of Brauner–Daugherty–Mason–Schilling and the singular Soergel bimodule polynomial-forcing of Elias–Ko–Libedinsky–Patimo, see below) must specialise to *some* statement about how residual symmetry organises length-colliding orbit elements. The $[2]_t$ appearance for $(2, 2, 0, 0)$ is the first instance of that; the second instance — where the residual acts by a group of order 3 or higher — is the diagnostic test for $(3, 3, 1, 0)$ -type cases. The trivial-stabiliser collision-and-asymmetry cases $(2, 1, 0)$ and $(3, 2, 1)$ are a third, distinct sub-regime in which the collision persists but no group-theoretic mechanism identifies the partners; that sub-regime is the least well-understood at present, and its appearance in the smallest possible non-chain examples is one of the surprises of the six-case data.

Mod- t is exact in all six cases. Setting $t = 0$ in the diagonal block collapses $(1 - t) \mapsto 1$; the sub-diagonal and above-diagonal t -shadows vanish. The classical Mason atom decomposition $B(\mu) = \bigsqcup_{\gamma} A_{\gamma}$ (Lascoux–Schützenberger 1990, Mason 2006) is recovered, and the Dim Lemma continues to require only this classical structure — it remains unconditional. In particular, the observation that the c_{γ} evaluated at $t = 0$ all equal 1 (both in the chain regime $[L + 1]_t|_{t=0} = 1$ and in the collision regime $[n]_t \cdot [k]_{tm}|_{t=0} = 1$) is not accidental: it is the statement that at the classical limit the Schur function s_{μ} decomposes as the sum of classical Demazure atoms indexed by the orbit, each with multiplicity one. The t -graded content of Route δ^+ lives entirely in the positive powers of t , where the various Gaussians distinguish themselves.

5 Categorical shadows

Two categorifications, both landed in the 2026-07-24-bis browse cycle, are candidate homes for a structural proof of Conjecture 8.

The crystal side: Brauner–Daugherty–Mason–Schilling 2607.12232. This paper gives a rigorous *classical* ($t = 0$) Bruhat contraction of the crystal skeleton on unions of Demazure crystals. Their tiles are the Young quasisymmetric Schur functions YQS_{α} , which are coarser than the Mason atoms A_{γ} that appear here (they collect multiple atoms into a single quasisymmetric block indexed by a composition rearrangement). The $t = 0$ specialisation of Route δ^+ ’s block-triangular matrix should therefore refine their tiled contraction. What is missing from their picture — and what any structural proof of δ^+ must supply — is the t -graded lift: the $[k]_{tm}$ factors that Conjecture 8(b) predicts do not appear anywhere in a classical crystal skeleton and must come from a t -graded refinement of the contraction.

The Soergel side: Elias–Ko–Libedinsky–Patimo 2407.13128. The Atomic Leibniz Rule is a polynomial-forcing statement for singular Soergel bimodules indexed by double-coset combinatorics $W_I \backslash W / W_J$. The name collision with the sprint’s nil-Hecke atoms is not (yet) a proved equivalence, but the shape of the singular double-coset table is precisely the residual-action combinatorics that

Conjecture 8(b) invokes. Reading 2407.13128 for whether its polynomial-forcing table produces $[k]_{t^m}$ factors of the shape observed here is a deep-read priority.

Neither framework has been read yet at the level of detail needed to convert either observation into a proof.

The Fock side (Robin's picture). The transfer-operator computations that Robin has been developing on the Fock side, in which $c_{\mu\nu}^\lambda$ arises through Kostka-matrix inversion, provide the fourth face of the polytope. On that side, the determinism of the Kostka inversion is the analogue of the uniqueness statement in the Dim Lemma; the Bruhat-chain vs. Bruhat-collision dichotomy should have a Fock-side avatar in the shape of the Kostka matrix relative to the residual action of $\text{Stab}(\mu)$. Whether the two dichotomies are literally the same statement in different notation, or are two features of a single underlying combinatorial object, is one of the sharper open questions left after these six cases.

The four faces of the conjectural polytope — Fock (Robin), atoms (Clio), crystals (Brauner–Daugherty–Mason–Schilling), Soergel bimodules (Elias–Ko–Libedinsky–Patimo) — each propose a different vocabulary for the same block-triangular structure. What Route δ^+ 's six-case data isolate is a minimum common statement that any of the four vocabularies must be able to prove: the chain-and-collision dichotomy on $\text{orbit}(\mu)$, with the specific coefficient values recorded in (2) and (3) above. If any one of the four frameworks can produce those coefficient values from its native combinatorics, Route δ^+ has found its naming home.

6 Next steps

1. *Bruhat-collision probes with larger stabilisers.* Compute c_γ for $(3, 3, 1, 0)$, $(3, 3, 2, 0)$, $(3, 2, 2, 0)$, and $(2, 2, 1, 1)$ and check whether $[k]_{t^m}$ factors appear with m predicted by the residual-orbit length. The $(2, 2, 1, 1)$ case, with stabiliser $\mathfrak{S}_2 \times \mathfrak{S}_2$, is the most stringent test.
2. *Structural proof of the Bruhat-chain case.* Attempt a direct proof of Conjecture 8(a) from the block-triangular structure alone. The Gaussian recursion $[L + 1]_t = 1 + t[L]_t$ should match the sub-diagonal t -shadow contribution to the recursion in L .
3. *Deep-read Elias–Ko–Libedinsky–Patimo 2407.13128* for singular double-coset polynomial forcing, checking whether the table produces the $[k]_{t^m}$ pattern seen at $(2, 2, 0, 0)$.

Remark 10 (On why $(2, 2, 1, 1)$ is the sharpest test). Compositions $(2, 2, 1, 1)$ has stabiliser $\mathfrak{S}_2 \times \mathfrak{S}_2$ of order 4; the orbit has size 6; Bruhat lengths on the orbit include a collision at length 1 (the two adjacent swaps of a 2-block against a 1-block). If the dichotomy holds and the residual action pairs those two length-1 elements into a single residual class of size 2, we should see a $[2]_{t^2}$ -type factor at $c_{(2,2,1,1)}$. If instead the stabiliser factor gives a $[k]_{t^m}$ with $m > 2$ (say $m = 4$ coming from the full stabiliser order), the residual-orbit-length formulation is wrong and only the raw stabiliser size matters. Either outcome is informative; the current data do not distinguish them.

What the six-case pattern reveals

The chain / collision dichotomy is not something we would have guessed from $(2, 2, 0)$ alone. It only becomes visible once one has in hand *both* a stabiliser-rich chain (like $(2, 2, 2, 0)$) and a stabiliser-poor collision (like $(2, 1, 0)$ or $(3, 2, 1)$): each isolates one of the two features that the earlier two-parameter Gaussian conjecture had accidentally conflated. What appears to matter is not the size of the stabiliser but whether Bruhat lengths collide on the orbit at all, and if so whether the residual

stabiliser action identifies the collision partners. The object under study — $c_\gamma(t)$ — is beginning to look like a genuinely two-story invariant: a Gaussian propagation along Bruhat chains, decorated by a residual-symmetry correction wherever the Bruhat order permits.